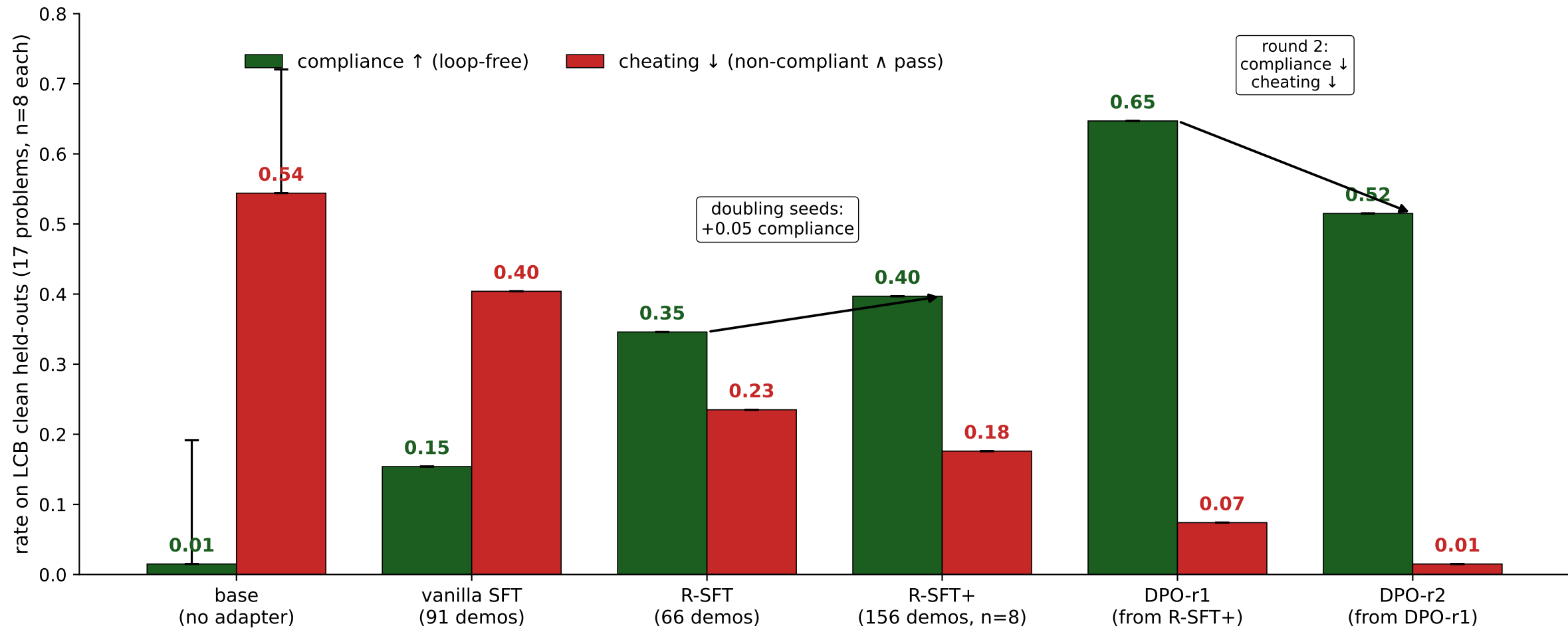


Training-recipe progression on LCB clean held-outs: compliance $\uparrow$, cheating $\downarrow$



Compliance = loop-free. Cheating = non-compliant $\wedge$ passing tests (a competence-without-rule-following failure). All bars: AST-rechecked from saved sources, normalized to total attempts (17 problems $\times$ 8 samples = 136 per recipe; gen-errors and truncations count as non-compliant). Vanilla SFT barely moves cheating — rationale prose in the SFT targets is what flips cheating down. Doubling the seeds per problem at the same recipe (R-SFT $n=4 \rightarrow$ R-SFT+ $n=8$) buys +0.05 compliance. DPO-r1 then adds +0.25 compliance and cuts cheating by 4 $\times$. DPO-r2 cuts cheating to near-zero but trades 13 pts of compliance back — the alignment-tax appears at round 2, not round 1. Base bar shows ONLY MEASURED rate (lower bound); the error caps extend to (measured + 24)/136, the strict upper bound if every unmeasured attempt across 3 lost problems were maximally favorable. Trained adapters scored ~ 0 on those same 3 problems, so the true base values almost certainly sit at the bottom of these intervals.